

① Find the general solution of the differential equation: $\frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$

② If a line makes an angle α, β, γ with the coordinate axes, then find the value of $\cos 2\alpha + \cos 2\beta + \cos 2\gamma$.

③ Events A and B are such that $P(A) = \frac{1}{2}$, $P(B) = \frac{7}{12}$ and $P(\bar{A} \cup \bar{B}) = \frac{1}{4}$

Find whether the events A and B are independent or not.

④ $\int_0^{\pi/4} \frac{1}{1 + \tan x} dx$

⑥ $\int \frac{2x}{(x^2+1)(x^2+2)} dx$

⑤ $\int e^x \cdot \sin 2x dx$

⑦ $\int_0^{2\pi} \frac{1}{1 + e^{\sin x}} dx$

⑧ Show that the lines are coplanar:

$$\frac{1-x}{2} = \frac{y-3}{4} = \frac{z}{-1} \quad \text{and} \quad \frac{x-4}{3} = \frac{2y-2}{-4} = \frac{z-1}{-1}$$

⑨ $\int \frac{x^2}{(x^2+1)(3x^2+4)} dx$

⑩ $\int_{-2}^1 \sqrt{5-4x-x^2} dx$

⑪ (Derivative) of $\log(\log(\log x^5))$ is : ? wrt x

⑫ The principal value of $\cos^{-1}(\frac{1}{2}) + \sin^{-1}(\frac{-1}{\sqrt{2}})$ is

⑬ If a matrix A is both symmetric and skew symmetric, then A is necessarily —